

AI in FinTech Presentation Script

Portfolio Optimisation and Interest Rate Pricing Using a Scoring Model

Fast speaking version: each member has enough text for about 5 minutes

Legend: RED = PowerPoint slide part | BLUE = Dashboard section part

Presentation Division

Member	Time	PowerPoint Part	Dashboard Sections
Marwa Zeineddine	5 min	Slide 1: Title + project overview	Overview
Ali Srour	5 min	Slide 2: Business problem + aim	Scorecard
Mhamad Naser Dine	5 min	Slide 3: Dataset and inputs	Client Explorer + Live Client Test
Lynn Khalil	5 min	Slide 4: Method summary	WOE & Binning + ROC & Validation
Maroun Mashaalany	5 min	Slide 5: Portfolio decision summary	Portfolio Risk + Accept / Reject + Final Decision
Ghandour Ghandour	5 min	Slide 6: Conclusion	Expected Loss + Risk Pricing + Model Performance + MATLAB & Excel Charts

Member 1 - Marwa Zeineddine

Opening, project overview, and dashboard overview

Target duration: about 5 minutes at medium-fast speaking speed

(Slide 1 - Title, course, and team introduction)

Good morning doctor. Today, we will present our project for the course AI in FinTech. Our project is titled Portfolio Optimisation and Interest Rate Pricing Using a Scoring Model. The objective of this project is to build a scorecard-based credit decision system that helps a bank evaluate loan applicants, estimate their probability of default, decide who should be accepted or rejected, and calculate the minimum interest rate that covers the expected credit risk.

The business context is simple but important. A bank has clients who are applying for loans. If the bank accepts everyone, it may suffer losses because some clients may default. If the bank rejects too many clients, it loses business opportunities and revenue. So the bank needs a balanced decision process. Our project helps the bank make this decision using data, not only judgment.

The dataset contains two parts. The first part is HistoricalData, which includes 500 previous bank clients. For these clients, we already know whether they defaulted or not. This historical data is used to build and validate the scoring model. The second part is ActualPortfolioData, which contains 20 current portfolio clients. These are the clients the bank wants to evaluate now. Each current client requests a loan of 100,000 dollars.

The model uses four predictors: Age, Income, Residential Status, and Employment Status. These variables describe the client profile and financial stability. The ID column is not used because it is only an identifier, and the existing PD column is not used as a predictor because including it would create data leakage. The model must learn the relationship between the client characteristics and the default outcome by itself.

Our final system produces five main outputs for the bank. First, it produces a score between 0 and 100 for each client. Second, it estimates the probability of default, or PD. Third, it gives an accept or reject decision. Fourth, it calculates the expected loss. Finally, it gives a minimum break-even interest rate for accepted clients.

The most important results are these: the model reaches an AUC of 0.6775, the KS statistic is 0.2892, and the optimal score cutoff is 48.2032. Using this cutoff, the model accepts 14 clients out of 20 and rejects 6 clients. The accepted exposure is 1,400,000 dollars because 14 accepted clients each request 100,000 dollars. The total expected loss of the accepted portfolio is 129,329.63 dollars, and the average minimum break-even annual interest rate is 9.24 percent.

(Dashboard - Overview section)

Now we move to the Overview section of the dashboard. This section summarizes the whole project visually. It shows the project workflow from data to model to portfolio decision and pricing. It is important to mention that MATLAB is the calculation engine of the project. MATLAB performs the scorecard workflow and produces the numerical outputs. The dashboard is the visualization and presentation layer. It helps us explain the results clearly and professionally.

In the Overview, we can explain the full chain of the project: client characteristics go into the scorecard model, the model produces a score and PD, the score is compared with the cutoff 48.2032, the client is accepted or rejected, then expected loss and risk-based pricing are calculated. This is the core idea of our project: data leads to score, score leads to risk, risk leads to decision, and decision leads to pricing.

So before going into the details, the main message is that our project connects machine learning, credit risk management, and banking pricing into one complete workflow. Now my colleague Ali will explain the business problem and the scorecard logic in more detail.

Member 2 - Ali Srou

Business problem, project aim, and scorecard interpretation

Target duration: about 5 minutes at medium-fast speaking speed

(Slide 2 - Business problem and project aim)

Now I will explain the business problem and the aim of the project. The bank faces a credit decision problem. It receives loan applications from clients, but it does not know with certainty which clients will repay and which clients may default. So the bank must answer two questions. First, should the bank accept or reject the client? Second, if the client is accepted, what minimum interest rate should the bank charge to compensate for credit risk?

This is a very important banking problem because credit risk directly affects profitability. If the bank accepts a risky client at a low interest rate, the expected loss may be higher than the expected income. This means the bank may lose money. On the other hand, if the bank rejects too many clients, it loses lending opportunities. So the bank needs a systematic method to classify risk and price it correctly.

The aim of our project is to build a credit scorecard model. A scorecard is useful because it gives each client a simple numerical score. A high score means the client is safer, and a low score means the client is riskier. The score helps the bank compare clients using one consistent scale instead of making decisions manually from many variables.

In this project, the score is scaled from 0 to 100. The safest possible profile receives a score close to 100, and the riskiest possible profile receives a score close to 0. For example, in our results, clients 4 and 5 receive a score of 100, which means they have the safest combination of characteristics. Client 1 receives a score of 0, which means his profile falls in the riskiest bins.

The score is then connected to the probability of default. The relationship is inverse: higher score means lower PD, and lower score means higher PD. This is exactly what we expect from a sound credit score. A score is easier for management to read, while PD gives the probability meaning behind the score.

The decision threshold in our project is 48.2032. This number was not chosen manually. It comes from the Kolmogorov-Smirnov statistic, which identifies the point where the model best separates good and bad clients. So the rule is: if the score is greater than or equal to 48.2032, the client is accepted. If the score is below 48.2032, the client is rejected.

(Dashboard - Scorecard section)

Now I will open the Scorecard section in the dashboard. This section shows how the client profile is converted into a credit score. The bank can use the score to rank clients from safest to riskiest. The scorecard also supports explainability, which is very important in banking. A bank cannot only say that a model rejected a client; it needs to understand which risk factors contributed to the score.

In our scorecard, the important predictors are Age, Income, Residential Status, and Employment Status. A more stable profile usually receives more positive points. For example, higher income, stable employment, home ownership, and older age are associated with safer credit profiles. Lower income, non-employment status, renting, and younger age tend to increase risk.

The scorecard is not a black-box model. It is transparent because it is based on binning, Weight of Evidence, and logistic regression. This is why scorecards are common in credit risk. They are not necessarily the most complex models, but they are interpretable and useful for banking management, risk committees, and reporting.

To summarize, the scorecard solves the first part of the business problem: it ranks clients by risk and supports the accept or reject decision. Later, the PD and expected loss solve the second part: the pricing of accepted clients. Now my colleague Mhamad will explain the dataset, portfolio clients, and live client testing.

Member 3 - Mhamad Naser Dine

Dataset, inputs, client explorer, and live testing

Target duration: about 5 minutes at medium-fast speaking speed

(Slide 3 - Dataset and inputs)

Now I will explain the dataset and the inputs used in the project. The project uses an Excel file called DataProjScoreCard.xlsx. This file contains two sheets. The first sheet is HistoricalData, and the second sheet is ActualPortfolioData. These two sheets have different roles in the project.

HistoricalData contains 500 past clients of the bank. For these clients, the default outcome is already known. The variable Default equals 0 for good clients who repaid their loan, and 1 for clients who defaulted. Because the outcome is known, this sheet is used to train and validate the scorecard model.

ActualPortfolioData contains 20 new applicants. For these 20 clients, the bank does not know the future default outcome. So the model applies what it learned from the 500 historical clients to these 20 new clients. The goal is to score them, estimate their PD, and decide who should be accepted or rejected.

It is important to distinguish between 20 clients and 4 predictors. The number 20 refers to the number of current clients in the portfolio. The number 4 refers to the four explanatory variables used by the model: Age, Income, Residential Status, and Employment Status. So we do not have 20 predictors; we have 20 clients and 4 predictors.

Age and Income are numeric variables. Residential Status and Employment Status are categorical variables. Residential Status can show whether the client is a renter or homeowner. Employment Status can show whether the client is employed or belongs to another category. These variables are relevant because they reflect repayment capacity and financial stability.

Each current portfolio client requests a loan of 100,000 dollars. This value is important because it is used later in the expected loss calculation. The project also assumes a recovery rate of 60 percent. This means that if a client defaults, the bank expects to recover 60 percent of the exposure. Therefore, the loss given default is 40 percent.

(Dashboard - Client Explorer section)

Now I will open the Client Explorer section in the dashboard. This section allows us to inspect the 20 portfolio clients. For each client, the dashboard shows important outputs such as score, PD, decision, expected loss, risk band, and minimum interest rate if accepted.

For example, client 4 has a score of 100 and a PD of 12.79 percent, so this client is very safe relative to the portfolio. Client 1 has a score of 0 and a PD of 70.59 percent, so this client is clearly risky and rejected. Client 8 has a score of 55.8354 and a PD of 33.50 percent. Client 8 is accepted because the score is above 48.2032, but he is still the riskiest accepted client, so he receives a high minimum rate of 13.40 percent.

This section is useful because it makes the model practical. Instead of only seeing formulas, the bank can look at each client and understand the decision. It also helps compare clients directly. If two clients request the same amount, the safer client should have a lower PD and lower minimum rate.

(Dashboard - Live Client Test section)

Now I will open the Live Client Test section. This section represents how the system could be used in practice. The bank can test a client profile and see how the model would evaluate it. The purpose is to demonstrate that the model can support future decision-making, not only analyze the fixed portfolio.

The key idea is that historical data teaches the model, and actual portfolio data receives the output. This separation is very important. We train the model on old clients where default is known, then apply it to new clients where default is unknown. This is the real use case of credit scoring.

To summarize, the dataset gives the model the information it needs. HistoricalData builds the model, ActualPortfolioData tests the current clients, and the dashboard makes the client-level results easy to understand. Now Lynn will explain the methodology: binning, WOE, logistic regression, and validation.

Member 4 - Lynn Khalil

Methodology, WOE, binning, logistic regression, and validation

Target duration: about 5 minutes at medium-fast speaking speed

(Slide 4 - Method summary)

Now I will explain the methodology used to build the scoring model. The workflow starts with data preparation, then binning, Weight of Evidence, logistic regression, scorecard scaling, and validation. The objective is to transform client characteristics into a risk score and a probability of default.

The first important step is binning. Binning means dividing a variable into groups. For example, age is not treated only as one continuous number. It is divided into age ranges. Income is also divided into income ranges. This helps the model understand how risk changes across different groups.

After binning, we calculate Weight of Evidence, or WOE. WOE measures how much a bin is associated with good clients or bad clients. A positive WOE means the bin contains more good clients relative to bad clients. A negative WOE means the bin is riskier because it contains more bad clients relative to good clients. This makes the variables more interpretable.

In our results, Income is the strongest predictor with Information Value 0.2355. Age is medium with IV 0.1879. Employment Status is also medium with IV 0.1143. Residential Status is weaker with IV 0.0417, but it still has enough predictive information to remain in the model. This makes economic sense because income and employment directly affect repayment ability.

After WOE transformation, MATLAB fits a logistic regression model. Logistic regression is suitable because the target variable is binary: a client either defaults or does not default. The output of logistic regression is the probability of default. All four predictors are statistically significant at the 5 percent level, so each predictor contributes to the model.

Then the scorecard points are created and scaled to a 0 to 100 range. The worst profile receives 0, and the best profile receives 100. This scaling makes the score easier to explain to management. Instead of talking only about model coefficients, we can talk about client scores.

(Dashboard - WOE & Binning section)

Now I will open the WOE & Binning section in the dashboard. This section visualizes how the predictors were grouped and how each group relates to default risk. For age, older applicants have safer profiles, with higher WOE. For residential status, homeowners are safer than renters. For employment status, employed clients are safer than others. For income, higher income is associated with lower default risk, and income is the strongest predictor.

This part is important because it explains the logic behind the model. The model is not making random decisions. It is using patterns found in the historical data. The dashboard helps show these patterns clearly.

(Dashboard - ROC & Validation section)

Now I will open the ROC & Validation section. The ROC curve measures how well the model separates good clients from bad clients. The AUC in our project is 0.6775. This is better than random, because a random model would have an AUC around 0.5. However, it is not excellent. It is a moderate but acceptable performance for a four-predictor academic scorecard trained on 500 observations.

The KS statistic is 0.2892, and the optimal KS score is 48.2032. This cutoff is used for the final accept or reject decision. It is important to be honest here: the model is useful, but not perfect. In a real bank, we would need out-of-sample validation, more variables, monitoring, and stress testing. But for this project, the complete scorecard workflow is correctly implemented.

To summarize, the methodology transforms raw client data into interpretable risk signals, fits a logistic model, creates a score, validates the model, and selects the cutoff. Now Maroun will explain how the score is used for portfolio selection and final decision-making.

Member 5 - Maroun Mashaalany

Portfolio risk, accept/reject decision, and final portfolio

Target duration: about 5 minutes at medium-fast speaking speed

(Slide 5 - Portfolio decision summary)

Now I will explain how the model is used to make the portfolio decision. After each client receives a score and probability of default, the bank needs to decide who should be accepted and who should be rejected. The decision rule is based on the optimal KS cutoff, which is 48.2032.

The rule is simple. If the client score is greater than or equal to 48.2032, the client is accepted. If the client score is below 48.2032, the client is rejected. This means that a score above the cutoff does not mean zero risk. It means that the risk is acceptable according to the model and the chosen cutoff. A score below the cutoff means the client is too risky for this portfolio.

Applying this rule to the 20 portfolio clients gives 14 accepted clients and 6 rejected clients. The accepted clients represent a total exposure of 1,400,000 dollars, because each accepted client asks for 100,000 dollars. The rejected clients are the ones with the lowest scores and highest probabilities of default.

For example, clients 4 and 5 are accepted with scores of 100 and PD of 12.79 percent. Client 6 is accepted with score 94.9267 and PD 14.45 percent. On the other side, client 1 is rejected with score 0 and PD 70.59 percent. Client 15 is rejected with score 4.7411 and PD 67.76 percent. These clients are rejected because their risk is much higher than the accepted group.

Client 8 is an important example. Client 8 has score 55.8354, which is above the cutoff, so he is accepted. But his PD is 33.50 percent, so he is not very safe. He is the riskiest accepted client. This shows the difference between accepted and safe. Accepted means the client passes the cutoff, but the interest rate must still reflect the risk.

(Dashboard - Portfolio Risk section)

Now I will open the Portfolio Risk section. This section shows the risk structure of the portfolio. The bank does not only care about individual clients. It also cares about the overall risk of the portfolio. If many risky clients are accepted, the portfolio may generate high expected losses. If only very safe clients are accepted, the portfolio may be too restrictive and lose business opportunities.

The portfolio risk section helps the bank see how risk is distributed across the 20 applicants. In our final selection, the accepted group contains low and medium risk clients, while the rejected clients fall in the high-risk group.

(Dashboard - Accept / Reject section)

Now I will open the Accept / Reject section. Here the decision is shown clearly for each client. The model accepts 14 and rejects 6. This is the main business output of the scoring model. The decision is systematic because it is based on the score cutoff and not on subjective judgment.

(Dashboard - Final Decision section)

Now I will open the Final Decision section. This section summarizes the final selected portfolio. The important result is that the model filters out the riskiest applicants while still keeping most of the portfolio. The rejection of 6 clients removes the most distressed applicants without being too conservative on the rest of the applicants.

To summarize, the portfolio decision is based on score, cutoff, and risk appetite. The model produces an actionable recommendation: lend to these clients, reject the others, then calculate expected loss and pricing for the accepted clients. Now Ghandour will explain the expected loss, risk pricing, model performance, and conclusion.

Member 6 - Ghandour Ghandour

Expected loss, risk pricing, model performance, and conclusion

Target duration: about 5 minutes at medium-fast speaking speed

(Slide 6 - Conclusion and financial outputs)

Now I will explain the financial part of the project: expected loss and minimum interest rate. After the model accepts clients, the bank must know how much risk cost each accepted loan creates. This is measured using expected loss.

Expected loss is calculated using the formula: Expected Loss equals PD multiplied by LGD multiplied by EAD. PD is the probability of default. LGD is loss given default. EAD is exposure at default, which is the loan amount. In our project, each loan amount is 100,000 dollars. The recovery rate is 60 percent, so LGD is 40 percent. This means if a client defaults, the bank expects to recover 60 percent and lose 40 percent.

So for every client, Expected Loss equals PD times 0.40 times 100,000. A safer client with low PD gives lower expected loss. A riskier client with high PD gives higher expected loss. For example, client 4 has PD 12.79 percent and expected loss around 5,114 dollars. Client 8 has PD 33.50 percent and expected loss around 13,402 dollars. Client 1 has PD 70.59 percent and expected loss around 28,234 dollars, but client 1 is rejected.

(Dashboard - Expected Loss section)

Now I will open the Expected Loss section. This section shows how much the bank expects to lose from each client on average. In our accepted portfolio, the total expected loss is 129,329.63 dollars. This is the amount the bank should provision against the accepted portfolio under the recovery rate assumption of 60 percent.

Expected loss is not the actual loss that will certainly happen. It is an average risk measure. Some clients may repay fully, and some may default. But expected loss helps the bank estimate the risk cost of the portfolio before making the loan decision.

(Dashboard - Risk Pricing section)

Now I will open the Risk Pricing section. The minimum break-even interest rate is calculated as PD multiplied by LGD. Since LGD is 40 percent, the rate is PD multiplied by 0.40. This rate covers the expected credit loss. If the bank charges below this rate, the expected interest income may be less than the expected loss. If the bank charges above this rate, the bank creates a positive expected margin.

For the accepted clients, the safest clients 4 and 5 have minimum interest rate 5.11 percent. Client 8, the riskiest accepted client, has minimum interest rate 13.40 percent. The average minimum break-even rate for the accepted portfolio is 9.24 percent. This is only the credit-risk floor. In a real bank, the final commercial rate should also include funding cost, operational cost, capital cost, profit margin, and market considerations.

(Dashboard - Model Performance section)

Now I will mention the Model Performance section. The AUC is 0.6775 and the KS statistic is 0.2892. These values show that the model has useful ranking power, but it is not perfect. We should not overclaim. The model is acceptable for the assignment and demonstrates the full credit scoring workflow, but a real banking deployment would require stronger validation.

(Dashboard - MATLAB & Excel Charts section)

Finally, the MATLAB & Excel Charts section shows that the project is reproducible. The Excel file provides the data, MATLAB produces the model outputs, and the dashboard visualizes those outputs. The MATLAB script writes tables such as unscaled points, scaled points, validation statistics, portfolio results, and accepted portfolio results.

To conclude, our project gives the bank a complete decision-support workflow. It starts with historical data, builds a scorecard model, scores 20 current clients, estimates PD, accepts 14 clients, rejects 6 clients, calculates total expected loss of 129,329.63 dollars, and gives an average minimum rate of 9.24 percent for the accepted portfolio. The strength of the project is that it does not stop at prediction. It connects prediction to decision, expected loss, and pricing. Thank you doctor. We are ready for your questions.